

CSC 380 — Quiz 04 Practice Guide

Start Here

Guide updated: 2026-09-24 afternoon (America/Phoenix) -- Start Here rewrite. Reviewed 2026-09-24 evening by Claude Opus 5.5 (see §8c).

If you only read one thing: the scope is confirmed but the due date is **not**, so the plan is to stay quiz-ready rather than count down. HW03 (due Tue Sep 29 11:59 PM) drills the same skills; finish it first. **Check Gradescope and D2L for a posted Quiz 04 due every Mon and Thu evening** (after 6:15 PM on Thu). When one appears, use the "short-notice" row below.

Next scheduled assessment

Field	Value	Label
Assessment	CSC 380 Quiz 04 (practice guide)	item announced (D2L Sep 18)
Due date/time	Needs verification -- not on Gradescope as of 2026-09-24 14:03 audit	NV -- do not invent
Coverage	Bayes' rule; law of total probability; independence; slides pp. 43-68	PUBLISHED SCOPE (confirmed)
Quiz policy	12 quizzes, 1.5% each; drop two lowest; no make-ups	confirmed (syllabus)
Past format	Quiz 01 = one rubric question, one page, 1.5 points	practice recommendation only

Do **not** schedule study during class blocks (Tue/Thu through Dec 9, Arizona time): CSC 252 9:30-10:45 AM; CSC 335 12:30-1:45 PM; CSC 345 2:00-3:15 PM; CSC 337 3:30-4:45 PM; CSC 380 5:00-6:15 PM. All times below are **Arizona (MST, UTC-7)**. Combined daily order: `weekly/2026-09-24-weekly-study-order.md`.

This is an open practice guide, not a countdown. The syllabus "quizzes every Tuesday" cadence is **not** a verified Quiz 04 due date.

Related (separate): CSC 380 **HW03** (written) + **HW03 Code** both due **Tue Sep 29 11:59 PM**. Overlap (Bayes/independence) is real; this guide does **not** solve HW3. After nearer exams (252 Test 2 Oct 1, 337 Oct 6, 335+345 Oct 8), interleave Quiz 04 mocks.

Topics to study first (order + WHY)

1. **380-Q04-refresh** -- **conditional probability** -- WHY: week-2 quiz was the weakest of the first three (words only); Bayes is just this rearranged.

2. **380-Q04-lotp** -- **law of total probability** -- WHY: published scope; it is the Bayes denominator; slides around p.18-20 of Probability 2 short, and the 43-68 window.
3. **380-Q04-bayes** -- **Bayes' rule + table method** -- WHY: published scope; starts at slide 43 in 26f380_probability2_short.pdf.
4. **380-Q04-indep** -- **independence vs mutually exclusive** -- WHY: published scope; product-rule check, not intuition.
5. **380-Q04-slides** -- **work the examples on slides 43-68** -- WHY: that page range is the only **confirmed** coverage.

Confirmed vs schedule-based: Scope is **confirmed**. Due datetime is **NV**. Do not build a day-before cram until Gradescope or D2L posts a due.

Day-by-day tasks (open practice; no invented due)

When	Window	Task	Guide / slides / problems	Min
Thu Sep 24	After 6:15 PM only	Do not invent a Friday quiz. If HW03 is open, work HW03 (not this key). Optional 15-min LOTP identity.	380-Q04-lotp; excerpt p.19; Qmu2	15
Fri Sep 25	After 345 HW2 / 252 deep	HW03 primary. Then one Bayes table.	excerpt p.47; T1 / S1	25
Sat Sep 26 - Sun Sep 27	After 252 / A4	HW03. One independence product check per day.	S2; CQ2	15
Mon Sep 28	Anytime	Aim to have HW03 written + Code nearly done tonight ; Tuesday's free time is mostly after 6:15 PM. One product check if time.	Quiz B (a)	HW03 + 10
Tue Sep 29	HW03 + HW03 Code due 11:59 PM. Class stack 9:30-6:15; work after 6:15 PM (or 10:45-12:30)	Submit both GS items. No Quiz 04 cram.	--	HW03
Wed Sep 30	No class	One 10-min mock (Quiz A) if HW03 is in.	Mock Quiz A	15
After Thu Oct 1 (252 Test 2 done)	Outside class blocks	Interleave mocks B, C, D and CQ1-CQ4.	Sec 4b + 4c	15-20 each
When a due date appears on GS/D2L (48+ hours of notice)	24-48 hours before that posted time	Light review of slides 43-68 + one mock.	380-Q04-slides	30

When	Window	Task	Guide / slides / problems	Min
When a due date appears with under 24 hours of notice	Next free window outside class blocks	Mock Quiz D + CQ4 (the two most exam-like), then check the keys. Skip everything else.	Sec 4b + 4c	20

On Tue/Thu, study only outside the whole 9:30-6:15 class stack (the free gaps are before 9:30, 10:45-12:30, and after 6:15). Deep drills Mon/Wed/Fri and weekends **after** nearer exam work.

Final self-check

- [] Mock Quizzes A-D each done in 10 minutes with at least 1.0 of 1.5
- [] CQ1-CQ4 done blind, then checked
- [] Three independence checks computed on both sides
- [] Slides 43-68 worked (Probability 2 short)
- [] **No countdown cram** until a due date is verified
- [] HW03 written + Code both submitted Sep 29

Slide excerpts (personal study channel only)

Excerpt 1 -- LOTP expansion (CSC 380, Probability 2 short, page 19)

Slide image (CSC 380 Probability 2 short p.19 LOTP) is not mirrored in this repo. It lives on the Munch box at `/home/box/shared/munch/study-guides/csc380/assessments/excerpts/csc380-prob2-p19-lotp.png`. The own-words caption below and the original deck page are enough to study from.

In my own words: if the B_i partition the sample space (no overlap, nothing left out), then $P(A) = \sum_i P(A | B_i) P(B_i)$. The slide's picture is "CS major" split across class years: the total CS-major probability is each year's CS rate times that year's share. This sum is the Bayes denominator.

Original deck: `/home/box/shared/munch/study-guides/_meta/d2l-harvest/csc380/csc380-26f380_probability2_short.pdf` (xinchenyu / D2L Resources), page 19. Personal study channel only. Quiz 04 published window is slides **43-68**; this p.19 page is the LOTP identity those later slides reuse.

Excerpt 2 -- Bayes formula (CSC 380, Probability 2 short, page 47)

Slide image (CSC 380 Probability 2 short p.47 Bayes) is not mirrored in this repo. It lives on the Munch box at `/home/box/shared/munch/study-guides/csc380/assessments/excerpts/csc380-prob2-p47-bayes.png`. The own-words caption below and the original deck page are enough to study from.

In my own words: $P(A \mid B) = P(A) P(B \mid A) / P(B)$. The easy derivation is the chain rule twice: $P(A \text{ and } B) = P(A) P(B \mid A) = P(B) P(A \mid B)$. The common exam mistake is swapping likelihood and posterior (saying $P(\text{disease} \mid +) = P(+ \mid \text{disease})$). Always compute $P(B)$ with LOTP first, or use the prior x likelihood table and divide by the column sum.

Original deck: same PDF, page 47 (inside the published 43-68 window). Personal study channel only.

Also work (no extra image): remaining examples on slides 48-53 and 43-68 of that deck; independence product checks in the long/short Probability 3 decks if you want more reps.

Banner — PUBLISHED SCOPE; due date NEEDS VERIFICATION. - Scope is published (D2L announcement, Sep 18, 2026): Bayes' rule, the law of total probability (LOTP), and independence, covering **slides pp. 43-68**. - **The due date is not verified**. Quiz 04 was not on Gradescope as of the 2026-09-24 audit (~14:03 Arizona time). The syllabus cadence ("quizzes every Tuesday unless noted") is **not** a verified due date. **Never invent a due date**. This is an open practice guide, not a countdown plan.

Canonical maps: `csc380-topic-map.md` * `announced-assessments-inventory-2026-09-24.md`

1. Header

Field	Value	Status
Course	CSC 380 Principles of Data Science (FA26), TuTh 5:00-6:15 PM, M Pacheco ILC 130	Confirmed
Assessment	Quiz 04 (practice guide)	Item announced
Due date/time	Needs verification — do not invent	NV
Coverage	Bayes' rule; LOTP; independence; slides 43-68	PUBLISHED SCOPE
Quiz policy (syllabus)	12 quizzes, each 1.5% of the grade; the two lowest are dropped; no make-ups	Confirmed (syllabus)
Past quiz format	Quiz 01 was one graded rubric question on a one-page submission, worth 1.5 points	Practice recommendation (Gradescope notes)
Guide updated	2026-09-24 (America/Phoenix) — Opus + Content-fold	

All times are **Arizona time (MST)**.

Sources (Content-pass 2026-09-24)

Source	Role
D2L announcement (Sep 18): Quiz 04	PUBLISHED SCOPE : Bayes, LOTP, independence; slides pp. 43-68
Content <code>Resources/quiz01_03.pdf</code>	Practice-pack grain (Quiz 01-03 style). Quiz 03 is LOTP+Bayes; used to shape Practice recommendation items below. Not a live quiz attempt.

Source	Role
Content Homework/homework02.pdf , homework03.pdf	Overlapping probability practice context (HW3 due Tue Sep 29, 11:59 PM). Do not treat this guide as HW solutions.
Content Homework/homework01.zip	Earlier pandas/data practice; not Quiz 04 core.
Gradescope format notes	One-page / 1.5-pt quiz habit (Quiz 01)

Related but separate: HW3 (Probability 2) is due **Tue Sep 29, 11:59 PM**. It has **two** Gradescope items, HW03 (written) and HW03 Code, and both must be submitted. It overlaps these topics, but HW3 is **not** this guide's target, and nothing here solves it.

Notation. In this guide, **not-A** means the complement of A. Slides often write it as A^c or as A with a bar over it. $P(A | B)$ is read "probability of A given B."

2. Topic checklist (learning order)

Order	Section ID	Topic	Blind Y	Check Y	Retake Y
1	380-Q04-refresh	Conditional probability refresher			
2	380-Q04-lotp	Law of total probability			
3	380-Q04-bayes	Bayes' rule			
4	380-Q04-indep	Independence (vs. mutually exclusive)			
5	380-Q04-slides	Work the examples on slides 43-68			

Alias: 380-L-prob3 $\sim$ 380-Q04-bayes , for calendar continuity.

3. Plain explanations + worked examples

Conditional probability refresher (380-Q04-refresh)

$P(A | B) = P(A \text{ and } B) / P(B)$, when $P(B) > 0$.

Rearranged, this gives the **multiplication rule**: $P(A \text{ and } B) = P(A | B) * P(B)$.

The Gradescope scores show the week-2 quiz (conditional probability) was the weakest of the three, so this refresher comes first.

Law of total probability (380-Q04-lotp)

If $B_1, \dots, B_k$ **partition** the sample space (they don't overlap and together cover everything):

$$P(A) = \sum_i P(A | B_i) * P(B_i)$$

With two parts: $P(A) = P(A | B)P(B) + P(A | \text{not-}B)P(\text{not-}B)$.

The intuition: split the world into cases, find A's probability in each case, and weight each case by how likely it is.

Bayes' rule (380-Q04-bayes)

$P(B_j | A) = P(A | B_j) \cdot P(B_j) / P(A)$, where $P(A)$ comes from LOTP.

Table method (it makes normalizing automatic):

Hypothesis	Prior	Likelihood P(data given H)	Prior x likelihood	Posterior
...	...	...	...	(row value) / (column sum)

Worked example (disease test). $P(D) = 0.01$, $P(+ | D) = 0.9$, and $P(+ | \text{not-}D) = 0.05$. - $P(+) = 0.9 \cdot 0.01 + 0.05 \cdot 0.99 = 0.009 + 0.0495 = 0.0585$. - $P(D | +) = 0.009 / 0.0585 \approx 0.154$.

The disease is rare (a low **base rate**), so most positive results are false positives, even though the test is fairly accurate.

Independence (380-Q04-indep)

A and B are independent when **$P(A \text{ and } B) = P(A) \cdot P(B)$** . Equivalently, $P(A | B) = P(A)$ (when $P(B) > 0$), or $P(B | A) = P(B)$.

- **Coin example.** For two fair flips, $P(H1 \text{ and } H2) = 1/4 = (1/2)(1/2)$, so the flips are independent.
- **Mutually exclusive is not independent.** If $P(A) > 0$ and $P(B) > 0$ but they cannot both happen, then $P(A \text{ and } B) = 0 \neq P(A) \cdot P(B)$. Knowing A happened tells you B did not, which is maximal dependence.
- To **check** independence, compute both sides of the product rule. Don't rely on intuition.

4. Practice questions (core)

Recall - R1. Write Bayes' rule for hypotheses H and not-H, given data D. - R2. State independence in three equivalent ways.

Solve - S1. Three urns are equally likely: U1 has 2 red and 1 blue, U2 has 1 red and 1 blue, U3 has 0 red and 2 blue. You pick an urn at random and draw one ball, which is red. What is $P(U1 | \text{red})$? - S2. $P(A) = 0.3$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.12$. Are A and B independent? - S3. $P(A | B) = 0.5$, $P(B) = 0.2$, $P(A | \text{not-}B) = 0.1$. Find $P(A)$ and $P(B | A)$.

Trace - T1. For S1, fill in the Bayes table: prior, likelihood, prior x likelihood, posterior.

Explain - E1. Why does LOTP appear in the denominator of Bayes' rule? - E2. What is the difference between independent and mutually exclusive?

Code (optional study aid) - C1. Simulate about 10,000 draws of the S1 setup in Python. Compare the fraction of red draws that came from U1 with the exact answer.

4b. One-page mock quizzes (1.5 points each) — Practice recommendation

Where this format comes from. Quiz 01's Gradescope result was one rubric question on a one-page submission, worth 1.5 points. Each mock below is **one multi-part problem that fits on one page**, with three parts worth 0.5 points each. The Quiz 04 interface and due date are **not known**. Give each mock **10 minutes**, closed-notes.

Mock Quiz A — LOTP + Bayes (factory)

Machine M1 makes 60% of the items, and 2% of its items are defective. Machine M2 makes the other 40%, and 5% of its items are defective. - (a) [0.5] Find $P(\text{defective})$. - (b) [0.5] An item is defective. Find $P(M2 \mid \text{defective})$. - (c) [0.5] In one sentence, explain why (b) is larger than 0.4.

Mock Quiz B — Independence (dice)

Roll two fair six-sided dice. Let $A = \text{"the first die shows 6"}$, $B = \text{"the sum is 7"}$, and $C = \text{"the sum is 12"}$. - (a) [0.5] Are A and B independent? Show the product check. - (b) [0.5] Are A and C independent? Show the product check. - (c) [0.5] $P(E) = 0.2$, $P(F) = 0.5$, and E and F are mutually exclusive. Are they independent?

Mock Quiz C — Table -> conditional -> independence

Of 200 students, 80 drink coffee. 50 of the coffee drinkers are night owls, and 40 of the non-coffee drinkers are night owls. - (a) [0.5] Find $P(\text{night owl})$. - (b) [0.5] Find $P(\text{coffee} \mid \text{night owl})$. - (c) [0.5] Are "coffee" and "night owl" independent? Justify with numbers.

Mock Quiz D — Bayes (spam filter)

$P(\text{spam}) = 0.3$. The word "free" appears in 40% of spam emails and in 5% of non-spam emails. - (a) [0.5] Find $P(\text{"free"})$. - (b) [0.5] Find $P(\text{spam} \mid \text{"free"})$. - (c) [0.5] Explain in one sentence the mistake in saying " $P(\text{spam} \mid \text{free}) = 0.4$ ".

Quick checks

- **Qmu1.** $P(A) = 0.5$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.1$. Independent? Give a one-line reason.
- **Qmu2.** Write LOTP for an event D with the partition $\{B, \text{not-}B\}$.
- **Qmu3.** $P(H) = 0.2$, $P(E \mid H) = 0.9$, $P(E \mid \text{not-}H) = 0.3$. Find $P(H \mid E)$, showing the LOTP denominator.
- **Qmu4.** True or false: mutually exclusive events with positive probabilities are independent.
- **Qmu5 (stretch).** Using the disease test from section 3, a second, independent test is also positive. Assume the two results are independent given disease status. Find $P(D \mid \text{both positive})$.

4c. Content-sourced similar-style practice — Practice recommendation

Where this comes from. Content Resources/quiz01_03.pdf is a practice pack (not a started quiz). Its Quiz 03 item is LOTP then Bayes on two groups with different rates — the same published Quiz 04 skills. The items below are **new** similar-style drills (numbers and story changed). Quiz 04 due date remains **NV**. Published scope is still Bayes / LOTP / independence only.

CQ1 — LOTP + Bayes (two clubs; Quiz 03 grain)

Two student clubs. Chess has 120 members; Debate has 80. Among Chess members, 25% always bring snacks to meetings. Among Debate members, 55% always bring snacks.

- (a) What fraction of all club members bring snacks?
- (b) A randomly selected student is bringing snacks. What is $P(\text{Debate} \mid \text{snacks})$?
- (c) In one sentence, why is (b) larger or smaller than Debate's membership share ($80/200 = 0.4$)?

CQ2 — Independence product check (HW3 grain)

Fair six-sided die. Let $A = \{2, 4, 6\}$ (even) and $B = \{1, 2, 3, 4\}$.

- (a) Compute $P(A)$, $P(B)$, and $P(A \text{ and } B)$. Are A and B independent?
- (b) Now let $A2 = \{2, 3, 4, 6\}$ and keep B. Are A2 and B independent?

CQ3 — Bayes (base-rate; HW3 shy-scientist grain)

At a large gathering, 70% of people are CS majors and 30% are data-science (DS) majors. 40% of CS majors own a mechanical keyboard; 80% of DS majors own one. You meet someone who owns a mechanical keyboard.

- (a) Find $P(\text{keyboard})$.
- (b) Find $P(\text{CS} \mid \text{keyboard})$.
- (c) Name the mistake in saying " $P(\text{CS} \mid \text{keyboard}) = 0.4$ ".

CQ4 — One-page mock (Quiz 01 format habit)

Factories F1 and F2 make 70% and 30% of chargers. Defect rates: 1% from F1, 4% from F2. A charger is defective.

- (a) [0.5] $P(\text{defective})$
- (b) [0.5] $P(\text{F2} \mid \text{defective})$
- (c) [0.5] Are "defective" and "from F2" independent? Justify with one product check (you may use (a)-(b)).

5. ANSWER KEY (separate — check after a blind attempt)

Core keys

R1. $P(H | D) = P(D | H)P(H) / [P(D | H)P(H) + P(D | \text{not-H})P(\text{not-H})]$

R2. $P(A \text{ and } B) = P(A)P(B)$; $P(A | B) = P(A)$; $P(B | A) = P(B)$. The last two require the conditioning event to have positive probability.

S1. $P(\text{red}) = (2/3)(1/3) + (1/2)(1/3) + 0 = 2/9 + 1/6 = 7/18$. So $P(U1 | \text{red}) = (2/9) / (7/18) = 4/7$.

S2. $0.3 \times 0.4 = 0.12$, which equals $P(A \text{ and } B)$. **Yes, independent.**

S3. $P(A) = 0.5 \times 0.2 + 0.1 \times 0.8 = 0.18$. $P(B | A) = 0.10 / 0.18 = 5/9$.

T1.

Urn	Prior	Likelihood	Prior x likelihood	Posterior
U1	1/3	2/3	$2/9 = 4/18$	4/7
U2	1/3	1/2	$1/6 = 3/18$	3/7
U3	1/3	0	0	0
Sum	1		7/18	1

E1. The denominator is $P(\text{data})$, the total probability of the data across all hypotheses. LOTP computes exactly that, and dividing by it makes the posteriors sum to 1.

E2. Mutually exclusive means $P(A \text{ and } B) = 0$. Independent means $P(A \text{ and } B) = P(A)P(B)$, which is usually greater than 0. With positive probabilities, the two properties cannot both hold.

C1. The empirical fraction should land near $4/7 \approx 0.571$, within about ± 0.02 for 10,000 draws.

Content-sourced keys (CQ1-CQ4)

CQ1. - (a) $P(\text{snacks}) = 0.25(120/200) + 0.55(80/200) = 0.25 \times 0.6 + 0.55 \times 0.4 = 0.15 + 0.22 =$

0.37. - (b) $P(\text{Debate} | \text{snacks}) = (0.55 \times 0.4) / 0.37 = 0.22 / 0.37 = 22/37 \approx 0.595^*$. - (c)

Debate members bring snacks at a higher rate, so conditioning on snacks shifts belief toward Debate (from 0.4 up to ~ 0.595).

CQ2. - (a) $P(A) = 3/6 = 1/2$. $P(B) = 4/6 = 2/3$. $P(A \text{ and } B) = P(\{2,4\}) = 2/6 = 1/3$. Product $(1/2)$

$(2/3) = 1/3$. *Independent.* - (b) $P(A2) = 4/6 = 2/3$. $P(A2 \text{ and } B) = P(\{2,3,4\}) = 3/6 = 1/2$. Product $(2/3)$
 $(2/3) = 4/9 \neq 1/2$. **Not independent.**

CQ3. - (a) $0.4 \times 0.7 + 0.8 \times 0.3 = 0.28 + 0.24 = 0.52$. - (b) $0.28 / 0.52 = 7/13 \approx 0.538$. - (c) 0.4 is $P(\text{keyboard} | \text{CS})$, the likelihood, not the posterior $P(\text{CS} | \text{keyboard})$.

CQ4. - (a) $0.70.01 + 0.30.04 = 0.007 + 0.012 = \mathbf{0.019}$. - (b) $0.012 / 0.019 = \mathbf{12/19} \approx \mathbf{0.632}$. - (c) $P(\text{defective and } F2) = 0.012$. $P(\text{defective})P(F2) = 0.0190.3 = 0.0057 \neq 0.012$. **Not independent.**

Mock quiz keys

Quiz A. - (a) $0.60.02 + 0.40.05 = 0.012 + 0.020 = \mathbf{0.032}$. - (b) $0.020 / 0.032 = \mathbf{0.625}$. - (c) M2 has the higher defect rate, so seeing a defect shifts belief toward M2, from the prior of 0.4 up to 0.625.

Quiz B. - (a) $P(A) = 1/6$. $P(B) = 6/36 = 1/6$. $P(A \text{ and } B) = P(\text{first die 6, second die 1}) = 1/36$, and $(1/6)(1/6) = 1/36$. **Independent.** - (b) $P(C) = 1/36$. $P(A \text{ and } C) = 1/36$, but $P(A)P(C) = 1/216$. *Not independent.* - (c) $P(E \text{ and } F) = 0$, but $P(E)P(F) = 0.1$. **Not independent.**

Quiz C. - (a) $(50 + 40) / 200 = \mathbf{0.45}$. - (b) $50 / 90 = \mathbf{5/9} \approx \mathbf{0.556}$. - (c) $P(\text{owl} | \text{coffee}) = 50/80 = 0.625$, which is not $P(\text{owl}) = 0.45$. Equivalently, $P(\text{coffee and owl}) = 0.25$, but $P(\text{coffee})P(\text{owl}) = 0.40.45 = 0.18$. **Not independent.**

Quiz D. - (a) $0.40.3 + 0.050.7 = 0.12 + 0.035 = \mathbf{0.155}$. - (b) $0.12 / 0.155 \approx \mathbf{0.774}$. - (c) 0.4 is $P(\text{free} | \text{spam})$, the likelihood. Treating it as $P(\text{spam} | \text{free})$, the posterior, ignores the prior and the base rate.

Quick-check keys - Qmu1. No. $0.50.4 = 0.2 \neq 0.1$. - **Qmu2.** $P(D) = P(D | B)P(B) + P(D | \text{not-}B)P(\text{not-}B)$. - **Qmu3.** $P(E) = 0.90.2 + 0.30.8 = 0.42$. $P(H | E) = 0.18 / 0.42 = 3/7 \approx 0.429$. - **Qmu4.** False. Their intersection has probability 0, but the product of their probabilities is greater than 0. - **Qmu5.** $0.810.01 / (0.810.01 + 0.00250.99) = 0.0081 / 0.010575 \approx \mathbf{0.766}$. Equivalently, use 0.154 as the new prior and update once more.

6. Common mistakes + readiness checklist

- Confusing $P(A | B)$ with $P(B | A)$ (base-rate neglect).
- Forgetting to normalize, i.e. skipping the LOTP denominator.
- Treating mutually exclusive events as independent.
- Using a "partition" whose parts overlap or leave gaps.
- **Inventing a Quiz 04 due date** from the Tuesday cadence.

Ready if: - [] Mock Quizzes A-D are each done in 10 minutes or less with at least 1.0 of 1.5 points - [] CQ1-CQ4 (Content-sourced) done blind, then checked - [] Three independence checks are done by computing both sides - [] The examples on slides 43-68 are worked - [] **No countdown cram** until the due date is verified

Start Here (top of this guide) is the canonical day-by-day. The later prep table is kept as a short copy.

7. Open practice checklist (no due-date cram plan)

Mode	Action
Anytime	Rotate short sets for 380-Q04-refresh, lotp, bayes, and indep
Alongside HW3 (due Sep 29, 11:59 PM)	Work these topics as separate practice. Upload both HW03 and HW03 Code.
When the due date is posted on Gradescope or D2L	Add a light review 24-48 hours before. Do not invent the date now.
Tuesdays and Thursdays	Keep light around classes. Do deep drills Mon/Wed/Fri and weekends.

8. Changes in this review (2026-09-24)

- Kept the **PUBLISHED SCOPE** and **due-date NV** banners unchanged in meaning.
- Replaced the complement symbol "c" with "not-A", because it rendered as a missing-glyph box in the old PDF. Added a notation note.
- Added quiz policy facts from the syllabus (12 quizzes, drop the 2 lowest, no make-ups) and the HW3 two-item submission reminder.
- Added Mock Quizzes A-D (one page, three parts at 0.5 points each), a filled Bayes table (T1), and a stretch sequential-update item. All have keys.
- Stated the weak-early-quiz signal in words only, with no scores (policy kept; this copy is now in the private begoneBOT/bob-jeff repo).
- Changed times to Arizona time.

8b. Content-fold (2026-09-24, America/Phoenix)

- Folded Content quiz01_03 + HW02/HW03 citations into Sources; added CQ1-CQ4 similar-style practice (**Practice recommendation**) from Quiz 03 / HW3 grain without copying live quiz or HW solutions.
- Published Quiz 04 scope unchanged (Bayes / LOTP / independence; slides 43-68). Due date still **NV**.
- ASCII-only math pass (no unicode subscripts / Sigma / mu) so PDF does not render missing-glyph boxes.

8c. Opus 5.5 review (2026-09-24 evening, America/Phoenix)

- Recomputed every key with exact fractions: S1-S3, T1, Mock Quizzes A-D, CQ1-CQ4, Qmu1-Qmu5, and the disease example. **No wrong answers found.** The C1 tolerance (+/-0.02) is about 2.5 standard errors for roughly 3,900 red draws, which is reasonable.
- Class-block fix:** Tue Sep 29 said only "no study 5:00-6:15". The whole 9:30-6:15 stack is off-limits. Added a Mon Sep 28 "HW03 nearly done" row so Tuesday is not a 6:15-11:59 PM scramble.
- Added a **short-notice** row (Mock D + CQ4, 20 min) for a due date posted with under 24 hours' notice, and a twice-weekly "check for a posted due" habit. **No due date was invented;** it is still NV.
- Fixed the garbled notation note ("not-A or not-A" is now " A^c or $\bar{A}$ ").

- Slide images are not in this repo, so the embeds became text pointers. The captions are kept.

9. Study-aid / AI-policy disclaimer

Study aid only — not for submitted work (Quiz 04, HW, exams). Follow CSC 380 and UA academic-integrity and AI policies.